# Topics in Causal Inference

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## 1 Week 1: Linear Regression as Covariate Adjustment and Balancing

### 1.1 Motivation

In empirical studies in the social and behavioral sciences, researchers are often interested in estimating the effect of a binary treatment or exposure on an outcome. Let:

- $Y$ denote an outcome (e.g., income),
- $Z \in \{0, 1\}$ denote a treatment (e.g., college degree),
- $X \in \mathbb{R}^p$ denote observed covariates (e.g., family background).

A naive comparison,

$$\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0],$$

generally fails to recover a "causal effect" when $Z$ and $X$ are statistically associated. This motivates *covariate adjustment*.

### 1.2 Linear Regression Adjustment

A standard approach is the linear regression model:

$$Y = \tau A + \beta^\top X + \varepsilon, \tag{1}$$

where $\mathbb{E}[\varepsilon \mid Z, X] = 0$ almost surely.

**Assume that the linear regression model is correctly specified.**

The coefficient $\tau$ is often interpreted as an *adjusted difference in means*, controlling linearly for covariates $X$.

$$\tau = \mathbb{E}\big[\mathbb{E}[Y \mid Z = 1, X] - \mathbb{E}[Y \mid Z = 0, X]\big].$$

### 1.3 Interpretation Without Causal Language

At this stage, we emphasize a purely statistical interpretation:

- Regression removes linear dependence of $Y$ on $X$.
- The remaining association between $Y$ and $Z$ is summarized by $\tau$.

Only later will we impose causal assumptions.

## 1.4 The Frisch–Waugh–Lovell (FWL) Theorem

**Theorem 1** (Frisch–Waugh–Lovell)**.** *Consider the linear regression of $Y$ on $(Z, X)$. The coefficient on $Z$ is numerically equal to the coefficient from the regression of $Y$ on $\tilde{Z}$, where:*

$$\tilde{Z} = Z - \mathbb{E}[Z \mid X].$$

*It is also numerically equal to the coefficient from the regression of $\tilde{Y}$ on $\tilde{Z}$, where:*

$$\tilde{Y} = Y - \mathbb{E}[Y \mid X],$$
$$\tilde{Z} = Z - \mathbb{E}[Z \mid X].$$

### 1.4.1 Geometric Interpretation

The FWL theorem states that:

- Both $Y$ and $Z$ are projected onto the orthogonal complement of the column space of $X$.
- The coefficient $\tau$ captures the linear relationship between the residualized variables.

Thus, regression compares variation in $Y$ and the part of $Z$ that is orthogonal to $X$.

## 1.5 Regression Coefficients as Weighted Averages

For simplicity, consider a binary treatment indicator $Z$.

Using FWL, the OLS estimator is:

$$\hat{\tau} = \frac{\sum_{i=1}^{n} \tilde{Z}_i Y_i}{\sum_{i=1}^{n} \tilde{Z}_i^2}. \tag{2}$$

This can be written as:

$$\hat{\tau} = \sum_{i=1}^{n} w_i Y_i = \sum_{i:Z_i=1} w_i Y_i - \sum_{i:Z_i=0} (-w_i) Y_i, \tag{3}$$

where:

$$w_i = \frac{\tilde{Z}_i}{\sum_{j=1}^{n} \tilde{Z}_j^2}.$$

**Balancing Between Treated and Control Groups**

Let $Z \in \{0, 1\}$ denote treatment and define the OLS weights

$$w_i = \frac{\tilde{Z}_i}{\sum_{j=1}^{n} \tilde{Z}_j^2}, \quad \tilde{Z}_i = Z_i - \hat{\mathbb{E}}[Z \mid X_i].$$

We first establish a global balance result and then derive balance between treated and control groups.

**Proposition 1** (Global Covariate Balance)**.**

$$\sum_{i=1}^{n} w_i X_i = 0.$$

*Proof.* By definition,

$$\sum_{i=1}^{n} w_i X_i = \frac{1}{\sum_{j=1}^{n} \tilde{Z}_j^2} \sum_{i=1}^{n} \tilde{Z}_i X_i.$$

Since $\tilde{Z}$ is the residual from the linear regression of $Z$ on $X$, the normal equations imply orthogonality:

$$\sum_{i=1}^{n} \tilde{Z}_i X_i = 0.$$

Substituting yields the result. □

We now decompose this balance by treatment status.

**Proposition 2** (Treated–Control Balance)**.** *The regression weights satisfy*

$$\sum_{i:Z_i=1} w_i X_i = \sum_{i:Z_i=0} (-w_i) X_i.$$

*Proof.* Partition the global sum by treatment status:

$$\sum_{i=1}^{n} w_i X_i = \sum_{i:Z_i=1} w_i X_i + \sum_{i:Z_i=0} w_i X_i.$$

From the previous proposition, the left-hand side equals zero. Therefore,

$$\sum_{i:Z_i=1} w_i X_i = -\sum_{i:Z_i=0} w_i X_i = \sum_{i:Z_i=0} (-w_i) X_i.$$

This equality shows that the weighted covariate total in the treated group equals the weighted covariate total in the control group when control weights are taken in absolute value. □

### Interpretation

Ordinary least squares implicitly constructs:

- a synthetic/reweighted treated group with weights $\{\omega_i\}$,
- a synthetic/reweighted control group with weights $\{-\omega_j\}$,

such that the covariate distributions are exactly balanced in sample.

**Conceptual takeaway.** *Regression adjustment works by reweighting observations so that treated and control groups are comparable with respect to observed covariates.*

This balancing interpretation is the point at which causal intuition enters: regression is not merely fitting a line, but constructing a comparison between groups that are comparable with respect to observed covariates.

The regression coefficient $\tau$ estimates the "causal effect" *only if* the linear specification is correctly specified. This raises an important question: what happens when it is not?

## 1.6 Example: When Linear Adjustment Fails

### Misspecified Linear Model: Excess Covariate Imbalance and Bias

Consider the data-generating process

$$Y_i = \tau Z_i + g(X_i) + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid Z_i, X_i] = 0,$$

where $g(\cdot)$ is an unknown nonlinear function.

Suppose we estimate $\tau$ using the misspecified linear regression

$$Y_i = \tau Z_i + \beta X_i + u_i,$$

where $g(X)$ is not linear in $X$.

Let $\tilde{Z}_i$ and $\tilde{X}_i$ denote residuals from regressing $Z$ on $X$ and an intercept, respectively.

By the Frisch–Waugh–Lovell theorem,

$$\hat{\tau} = \frac{\sum_{i=1}^n \tilde{Z}_i Y_i}{\sum_{i=1}^n \tilde{Z}_i^2}.$$

Substitute the true outcome model:

$$\hat{\tau} = \tau + \frac{\sum_{i=1}^n \tilde{Z}_i g(X_i)}{\sum_{i=1}^n \tilde{Z}_i^2} + \frac{\sum_{i=1}^n \tilde{Z}_i \varepsilon_i}{\sum_{i=1}^n \tilde{Z}_i^2}.$$

Taking expectations conditional on $X, Z$,

$$\mathbb{E}[\hat{\tau} \mid X, Z] = \tau + \frac{\sum_{i=1}^n \tilde{Z}_i g(X_i)}{\sum_{i=1}^n \tilde{Z}_i^2}.$$

### Bias in the Treatment Effect Estimate

The finite-sample bias is therefore

$$\boxed{\mathrm{Bias}(\hat{\tau} \mid X, Z) = \frac{\sum_{i=1}^n \tilde{Z}_i g(X_i)}{\sum_{i=1}^n \tilde{Z}_i^2}}$$

This bias vanishes if and only if

$$\sum_{i=1}^n \tilde{Z}_i g(X_i) = 0.$$

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### Excess Covariate Imbalance

Define the regression weights

$$w_i := \frac{\tilde{Z}_i}{\sum_{j=1}^n \tilde{Z}_j^2}.$$

The weighted covariate imbalance induced by the regression is

$$\sum_{i=1}^n w_i X_i = 0,$$

but for nonlinear functions $g(X)$,

$$\sum_{i=1}^{n} w_i g(X_i) \neq 0.$$

Define the *excess imbalance* for $g(X)$ as

$$\boxed{\Delta_g := \sum_{i:Z_i=1} w_i g(X_i) - \sum_{i:Z_i=0} (-w_i) g(X_i) = \sum_{i=1}^{n} w_i g(X_i).}$$

Then the bias can be written compactly as

$$\boxed{\mathrm{Bias}(\hat{\tau} \mid X, Z) = \Delta_g.}$$

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**Interpretation**

- Linear regression exactly balances covariates only in the span of the included regressors.
- Misspecification induces imbalance in functions of $X$ not included in the model.
- Bias arises from the failure to balance $g(X)$, not from imbalance in $X$ itself.

This illustrates that regression adjustment is a *balancing method*, but only for the covariate space it models.

### 1.7 Key limitations

Linear regression assumes:
linearity
constant effects
no extrapolation
Thus, if any of these assumptions do not hold, i.e., the model is not correctly specified, we cannot recover the "causal effect" of $Z$ on $Y$ using a linear regression.

There are other challenges as well. For example, OLS balancing is global, not exact. Also, Poor overlap $\rightarrow$ extreme weights

# APPENDIX

## Proof of the Frisch–Waugh–Lovell (FWL) Theorem

**Theorem 2** (Frisch–Waugh–Lovell)**.** *Consider the linear regression of $Y$ on $(Z, X)$, where $Z$ is a scalar regressor and $X$ may be a vector of covariates. Let*

$$\tilde{Z} = Z - E[Z \mid X], \qquad \tilde{Y} = Y - E[Y \mid X].$$

*The coefficient on $Z$ in the linear regression of $Y$ on $(Z, X)$ is numerically equal to:*

1. *the coefficient from the linear regression of $Y$ on $\tilde{Z}$, and*
2. *the coefficient from the linear regression of $\tilde{Y}$ on $\tilde{Z}$.*

## Proof

We proceed in steps.

**Step 1: The population linear projection.** Consider the population linear regression of $Y$ on $(Z, X)$. This is defined as the solution $(\beta, \gamma)$ to the minimization problem

$$\min_{\beta,\gamma} \mathrm{E}\Big[(Y - \beta Z - \gamma^\top X)^2\Big].$$

The normal equations for this problem are

$$\mathrm{E}\Big[Z(Y - \beta Z - \gamma^\top X)\Big] = 0, \tag{4}$$

$$\mathrm{E}\Big[X(Y - \beta Z - \gamma^\top X)\Big] = 0. \tag{5}$$

Our goal is to characterize the coefficient $\beta$.

**Step 2: Partialling out $X$.** Define the residuals from projecting $Z$ and $Y$ onto $X$:

$$\tilde{Z} = Z - \mathrm{E}[Z \mid X], \qquad \tilde{Y} = Y - \mathrm{E}[Y \mid X].$$

By construction,

$$\mathrm{E}[\tilde{Z} \mid X] = 0, \qquad \mathrm{E}[\tilde{Y} \mid X] = 0,$$

and hence

$$\mathrm{E}[X\tilde{Z}] = 0, \qquad \mathrm{E}[X\tilde{Y}] = 0.$$

**Step 3: Using the normal equations.** From equation (5), the coefficient $\gamma$ satisfies

$$\mathrm{E}[XY] = \beta \mathrm{E}[XZ] + \mathrm{E}[XX^\top]\gamma.$$

This implies that $\mathrm{E}[Y \mid X]$ is the linear projection of $Y$ onto $X$ given $(\beta, Z)$, and similarly for $\mathrm{E}[Z \mid X]$.

Substitute the decompositions

$$Y = \tilde{Y} + \mathrm{E}[Y \mid X], \qquad Z = \tilde{Z} + \mathrm{E}[Z \mid X]$$

into the first normal equation (4):

$$\mathrm{E}\Big[(\tilde{Z} + \mathrm{E}[Z \mid X])\big(\tilde{Y} + \mathrm{E}[Y \mid X] - \beta(\tilde{Z} + \mathrm{E}[Z \mid X]) - \gamma^\top X\big)\Big] = 0.$$

Using $\mathrm{E}[\tilde{Z} \mid X] = 0$ and $\mathrm{E}[\tilde{Y} \mid X] = 0$, all terms involving $X$ drop out, and the condition simplifies to

$$\mathrm{E}[\tilde{Z}(\tilde{Y} - \beta\tilde{Z})] = 0.$$

**Step 4: Identification of $\beta$.** The last equation is exactly the normal equation for the linear regression of $\tilde{Y}$ on $\tilde{Z}$. Therefore,

$$\beta = \frac{\mathrm{E}[\tilde{Z}\tilde{Y}]}{\mathrm{E}[\tilde{Z}^2]}.$$

This is the coefficient from:

- the regression of $\tilde{Y}$ on $\tilde{Z}$, and
- equivalently, the regression of $Y$ on $\tilde{Z}$, since $\mathrm{E}[Y \mid X]$ is orthogonal to $\tilde{Z}$.

**Conclusion.** Thus, the coefficient on $Z$ in the regression of $Y$ on $(Z, X)$ is numerically equal to the coefficient obtained by first partialling out $X$ from $Z$ (and optionally from $Y$) and then regressing the resulting residuals. This completes the proof. $\square$

# Homework Problems

## Week 1: Linear Regression, Covariate Adjustment, and FWL

**Problem 1** (FWL with Multivariate Treatment)**.** *Let $Y \in \mathbb{R}^n$ be an outcome vector, $Z \in \mathbb{R}^{n \times q}$ a matrix of treatment variables, and $X \in \mathbb{R}^{n \times p}$ a matrix of covariates. Consider the linear regression*

$$Y = Z\beta + X\gamma + \varepsilon,$$

*estimated by ordinary least squares.*

1. *Define the residual-maker matrix*

$$M_X = I - X(X^\top X)^{-1}X^\top.$$

   *Show that the OLS estimator $\hat{\beta}$ satisfies*

$$\hat{\beta} = (Z^\top M_X Z)^{-1} Z^\top M_X Y.$$

2. *Let $\tilde{Z} = M_X Z$ denote the residuals from regressing each column of $Z$ on $X$. Show that $\hat{\beta}$ is numerically equal to the coefficient vector obtained from regressing $Y$ on $\tilde{Z}$.*

3. *Interpret this result in words. What does it say about how regression "adjusts" for covariates?*

**Problem 2** (Misspecification and Orthogonality)**.** *Suppose the true data-generating process is*

$$Y = \tau Z + f(X) + \varepsilon,$$

*where $f(X)$ is an unknown (possibly nonlinear) function of $X$ and $E[\varepsilon \mid Z, X] = 0$. You instead estimate the misspecified simple linear model*

$$Y = \alpha + \beta Z + X^\top \gamma + u.$$

1. *Assume that $Z$ is mean-independent of $X$, i.e.,*

$$E[Z \mid X] = E[Z].$$

   *Show that $\hat{\beta}$ is consistent for $\tau$.*

2. *Explain why $\hat{\beta}$ may not be unbiased for $\tau$.*

3. *Explain why the result in 1. does* not *contradict the general dangers of model misspecification.*

4. *Give an example in which $Z$ is correlated with $X$ and explain why $\hat{\beta}$ would generally be biased in this case.*